Department of Mathematics, IIT Madras
MA2101 Linear Algebra via Matrices
Assignment 1

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1. For each of the following system of linear equations, find the number of solutions geometrically:

   (a) $x_1 + 2x_2 = 4, \quad -2x_1 - 4x_2 = 4$
   (b) $-x_1 + 2x_2 = 3, \quad 2x_1 - 4x_2 = -6$
   (c) $x_1 + 2x_2 = 1, \quad x_1 - 2x_2 = 1, \quad -x_1 + 6x_2 = 3$

   Answer: (a) No solution, (b) Infinite solutions, (c) Unique solution.

2. Show that the system of linear equations $a_1x_1 + x_2 = b_1,\ a_2x_1 + x_2 = b_2$ has a unique solution if $a_1 \neq a_2$. Is the converse true? **Answer:** Converse is true.

3. Compute $AB,\ CA,\ DC,\ DCAB,\ A^2,\ D^2$ and A^3B^2, where

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & -1 \\ 4 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} -1 & 2 \\ 2 & -1 \\ 1 & 3 \end{bmatrix}, \quad D = \begin{bmatrix} 3 & 2 & 1 \\ 4 & -6 & 0 \\ 1 & -2 & -2 \end{bmatrix}.$$

   Answer: $AB = \begin{bmatrix} 20 & -2 \\ 12 & -1 \end{bmatrix}, \quad CA = \begin{bmatrix} 0 & 1 \\ 3 & 4 \\ 5 & 9 \end{bmatrix}, \quad DC = \begin{bmatrix} 2 & 7 \\ -16 & 14 \\ -7 & -2 \end{bmatrix}, \quad DCAB = \begin{bmatrix} 124 & -11 \\ -152 & 18 \\ -164 & 16 \end{bmatrix},$

$$A^2 = \begin{bmatrix} 7 & 12 \\ 4 & 7 \end{bmatrix}, \quad D^2 = \begin{bmatrix} 18 & -8 & 1 \\ -12 & 44 & 4 \\ -7 & 18 & 5 \end{bmatrix}, \quad A^3B^2 = \begin{bmatrix} 1032 & -284 \\ 596 & -164 \end{bmatrix}$$

4. Let $A = [a_{ij}] \in \mathbb{F}^{2\times 2}$ with $a_{11} \neq 0$. Let $b = a_{21}/a_{11}$. Show that there exists $c \in \mathbb{F}$ such that $A = \begin{bmatrix} 1 & 0 \\ b & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ 0 & c \end{bmatrix}$. What could be c? **Answer:** $c = (a_{11}a_{22} - a_{12}a_{21})/a_{11}$

5. Let $A \in \mathbb{F}^{m\times n}$ and let $B \in \mathbb{F}^{n\times k}$. For $1 \leq i \leq m,\ 1 \leq j \leq k$, show that
   (a) $(AB)_{i\star} = A_{i\star}\, B$ (b) $(AB)_{\star j} = A\, B_{\star j}$

6. Construct two 3×3 matrices A and B such that $AB = 0$ but $BA \neq 0$.

   Answer: Consider the following 3×3 matrices: $A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$

7. Can you construct invertible 2×2 matrices A and B such that $AB = 0$? **Answer:** Not Possible.

8. Let $A, B \in \mathbb{F}^{n\times n}$ be such that $AB = 0$. Then which of the following is/are true, and why?
   (a) At least one of A or B is the zero matrix.
   (b) At least one of A or B is invertible.
   (c) At least one of A or B is non-invertible.
   (d) If $A \neq 0$ and $B \neq 0$, then neither is invertible. **Answer:** (a) False, (b) False, (c) True, (d) True

9. Let A be the 4×4 matrix whose super diagonal entries are all 1, and all other entries 0. Show that $A^n = 0$ for $n \geq 4$.

10. Let A be the 4×4 matrix with each diagonal entry as $\frac{1}{2}$, and each non-diagonal entry as $-\frac{1}{2}$. Compute A^n for $n \in \mathbb{N}$. **Answer:** $A^n = A$ (if n is odd), and $A^n = I$ (if n is even)

11. Determine A^t, $\overline{A}$, A^*, A^*A and AA^*, where

(a) $A = \begin{bmatrix} -1 & 2 & 3 & 1 \\ 2 & -1 & 0 & 3 \\ 0 & -1 & -3 & 1 \end{bmatrix}$ (b) $A = \begin{bmatrix} 1 & -2+i & 3-i \\ i & -1-i & 2i \\ 1+3i & -i & -3 \\ -2 & 0 & -i \end{bmatrix}$

Answer:

(a) $A^t = \begin{bmatrix} -1 & 2 & 0 \\ 2 & -1 & -1 \\ 3 & 0 & -3 \\ 1 & 3 & 1 \end{bmatrix}$, $\overline{A} = A$, $A^* = A^t$, $A^*A = \begin{bmatrix} 5 & -4 & -3 & 5 \\ -4 & 6 & 9 & -2 \\ -3 & 9 & 18 & 0 \\ 5 & -2 & 0 & 11 \end{bmatrix}$, $AA^* = \begin{bmatrix} 15 & -1 & -10 \\ -1 & 14 & 4 \\ -10 & 4 & 11 \end{bmatrix}$.

(b) A^t, $\overline{A}$, A^* are trivial. Now,

$$A^*A = \begin{bmatrix} 16 & -6+i & 2+10i \\ -6-i & 8 & -9-6i \\ 2-10i & -9+6i & 24 \end{bmatrix} \quad \text{and} \quad AA^* = \begin{bmatrix} 16 & -1-10i & -9-2i & -1+3i \\ -1+10i & 7 & 4-6i & -2-2i \\ -9+2i & 4+6i & 20 & -2-9i \\ -1-3i & -2+2i & -2+9i & 5 \end{bmatrix}.$$

12. Let $A \in \mathbb{C}^{m\times n}$. Suppose $AA^* = I_m$. Does it follow that $A^*A = I_n$?

13. Show the following:

(a) $E_{ij}E_{jm} = E_{im}$; and if $j \neq k$, then $E_{ij}E_{km} = 0$.

(b) Each $A = [a_{ij}] \in \mathbb{F}^{n\times n}$ can be written as $A = \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}E_{ij}$.

14. Compute $E[2,3]A$, $E_i[2]A$, $E_{-1/2}[1,3]A$ and $E_i[1,2]A$, where

(a) $A = \begin{bmatrix} -1 & 2 & 3 & 1 \\ 2 & -1 & 0 & 3 \\ 0 & -1 & -3 & 1 \end{bmatrix}$ (b) $A = \begin{bmatrix} 1 & -2+i & 3-i \\ i & -1-i & 2i \\ 1+3i & -i & -3 \\ -2 & 0 & -i \end{bmatrix}$

Answer:

(a) $E[2,3]A = \begin{bmatrix} -1 & 2 & 3 & 1 \\ 0 & -1 & -3 & 1 \\ 2 & -1 & 0 & 3 \end{bmatrix}$, $E_i[2]A = \begin{bmatrix} -1 & 2 & 3 & 1 \\ 2i & -i & 0 & 3i \\ 0 & -1 & -3 & 1 \end{bmatrix}$

$E_{-1/2}[1,3]A = \begin{bmatrix} -1 & 2.5 & 4.5 & 0.5 \\ 2 & -1 & 0 & 3 \\ 0 & -1 & -3 & 1 \end{bmatrix}$, $E_i[1,2]A = \begin{bmatrix} -1+2i & 2-i & 3 & 1+3i \\ 2 & -1 & 0 & 3 \\ 0 & -1 & -3 & 1 \end{bmatrix}$

(b) $E[2,3]A = \begin{bmatrix} 1 & -2+i & 3-i \\ 1+3i & -i & -3 \\ i & -1-i & 2i \\ -2 & 0 & -i \end{bmatrix}$, $E_i[2]A = \begin{bmatrix} 1 & -2+i & 3-i \\ -1 & 1-i & -2 \\ 1+3i & -i & -3 \\ -2 & 0 & -i \end{bmatrix}$

$E_{-1/2}[1,3]A = \begin{bmatrix} 0.5-1.5i & -2+1.5i & 4.5-i \\ i & -1-i & 2i \\ 1+3i & -i & -3 \\ -2 & 0 & -i \end{bmatrix}$, $E_i[1,2]A = \begin{bmatrix} 1+i & -1 & 1-i \\ i & -1-i & 2i \\ 1+3i & -i & -3 \\ -2 & 0 & -i \end{bmatrix}$

15. Let α be a nonzero scalar. Show the following:

(a) $(E[i,j])^t = E[i,j]$, $(E_\alpha[i])^t = E_\alpha[i]$, $(E_\alpha[i,j])^t = E[j,i]$.

(b) $(E[i,j])^{-1} = E[i,j]$, $(E_\alpha[i])^{-1} = E_{1/\alpha}[i]$, $(E_\alpha[i,j])^{-1} = E_{-\alpha}[i,j]$.

16. For each of the following pairs of matrices, find an elementary matrix E such that $B = EA$.

(a) $A = \begin{bmatrix} 2 & 1 & 3 \\ 3 & 1 & 4 \\ -2 & 4 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 & 3 \\ -2 & 4 & 5 \\ 1 & 5 & 9 \end{bmatrix}$

(b) $A = \begin{bmatrix} 4 & -2 & 3 \\ 1 & 0 & 2 \\ 0 & 3 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 3 & -2 & 1 \\ 1 & 0 & 2 \\ 0 & 3 & 5 \end{bmatrix}$

Answer: (a) $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$, (b) $E = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

17. For each of the following pairs of matrices, find an elementary matrix E such that $B = AE$. [Hint: The requirement is $B^t = E^t A^t$.]

(a) $A = \begin{bmatrix} 3 & 1 & 4 \\ 4 & 1 & 2 \\ 2 & 3 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 1 & 3 \\ 2 & 1 & 4 \\ 1 & 3 & 2 \end{bmatrix}$

(b) $A = \begin{bmatrix} 2 & -2 & 3 \\ -1 & 4 & 2 \\ 3 & 1 & -2 \end{bmatrix}$, $B = \begin{bmatrix} 4 & -2 & 3 \\ -2 & 4 & 2 \\ 6 & 1 & -2 \end{bmatrix}$

Answer: (a) $E = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$, (b) $E = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

18. Which of the following matrices are in RREF, and which are not?

(a) $\begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$.

Answer: (a) No (b) Yes (c) Yes

19. Compute the RREF of the following matrices:

(a) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 2 & -1 & -1 & 0 \\ 1 & 0 & 1 & -4 \\ 0 & 1 & -1 & -4 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 2 & 1 & -1 \\ 0 & 2 & 3 & 3 \\ 1 & -1 & -3 & -4 \\ 1 & 1 & 5 & -2 \end{bmatrix}$

Answer: (a) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, (b) $\begin{bmatrix} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & -5 \\ 0 & 0 & 1 & -1 \end{bmatrix}$, (c) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$.

20. Suppose the RREF of a matrix A is equal to $\begin{bmatrix} 1 & 2 & 0 & 3 & 1 & -2 \\ 0 & 0 & 1 & 2 & 4 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.

If $A_{\star 1} = [1,\ 1,\ 3,\ 4]^t$ and $A_{\star 3} = [2,\ -1,\ 1,\ 3]^t$, then determine $A_{\star 6}$. **Answer:** $A_{\star 6} = [8, -7, -1, 7]^t$

21. Consider the row vectors $v_1 = [1, 2, 3, 4]$, $v_2 = [2, 0, 1, 1]$, $v_3 = [-3, 2, 1, 2]$, and $v_4 = [1, -2, -2, -3]$. Construct a row vector $v \in \mathbb{R}^{4\times 1}$ which is not expressible as $av_1 + bv_2 + cv_3 + dv_4$ for any $a, b, c, d \in \mathbb{R}$. [Hint: Compute the RREF of $A = [v_1^t \ \ v_2^t \ \ v_3^t \ \ v_4^t]$.]

22. For each $n > 2$, construct an $n \times n$ nonzero matrix where no row is a scalar multiple of another row but its determinant is 0.

23. Compute A^{-1} using $\text{adj}(A)$, where $A = \begin{bmatrix} 1 & 0 & 0 & 1 \\ -1 & 1 & 0 & 1 \\ -1 & -1 & 1 & 1 \\ -1 & -1 & -1 & 1 \end{bmatrix}$.

Answer: $A^{-1} = \begin{bmatrix} 1/2 & -1/4 & -1/8 & -1/8 \\ 0 & 1/2 & -1/4 & -1/4 \\ 0 & 0 & 1/2 & -1/2 \\ 1/2 & 1/4 & 1/8 & 1/8 \end{bmatrix}$.

24. Let $A, B \in \mathbb{F}^{3\times 3}$ with $\det(A) = 2$ and $\det(B) = 3$. Determine
(a) $\det(4A)$ (b) $\det(AB)$ (c) $\det(5AB)$ (d) $\det(2A^{-1}B)$

Answer: (a) 128 (b) 6 (c) 750 (d) 12

25. Let $A, B \in \mathbb{F}^{2\times 2}$ and let $E = [e_{ij}]$ with $e_{11} = e_{22} = 0$, $e_{12} = e_{21} = 1$. Show that if $B = EA$, then $\det(A + B) = \det(A) + \det(B)$.

26. Give examples of A and B in $\mathbb{R}^{2\times 2}$ so that $\det(A + B) \neq \det(A) + \det(B)$.

Answer: Take $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$

27. Compute the inverses of the following matrices, if possible:

(a) $\begin{bmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ -1 & 1 & 2 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 4 & -6 \\ -1 & -1 & 3 \\ 1 & -2 & 3 \end{bmatrix}$ (c) $\begin{bmatrix} 3 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 \\ 1 & 1 & 2 & -1 \\ -2 & 1 & -1 & 3 \end{bmatrix}$

Answer: (a) $\begin{bmatrix} \frac{2}{5} & \frac{4}{15} & -\frac{1}{3} \\ -\frac{1}{5} & \frac{2}{5} & 0 \\ \frac{11}{15} & -1 & \frac{1}{3} \end{bmatrix}$ (b) $\begin{bmatrix} \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{2}{3} & 1 & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} & \frac{1}{3} \end{bmatrix}$ (c) $\begin{bmatrix} \frac{7}{36} & \frac{5}{36} & \frac{-2}{9} & \frac{-1}{4} \\ \frac{-2}{9} & \frac{5}{9} & \frac{1}{9} & 0 \\ \frac{5}{36} & \frac{-17}{36} & \frac{5}{9} & \frac{1}{4} \\ \frac{1}{4} & \frac{-1}{4} & 0 & \frac{1}{4} \end{bmatrix}$

28. Let $A = \begin{bmatrix} 2 & 1 \\ 6 & 4 \end{bmatrix}$. Express A and A^{-1} as products of elementary matrices.

Answer: (a) $A = E_1E_2E_3$, (b) $A^{-1} = E_3E_2E_1$, where $E_1 = \begin{bmatrix} 1/2 & 0 \\ 0 & 1 \end{bmatrix}, E_2 = \begin{bmatrix} 1 & 0 \\ -6 & 1 \end{bmatrix}, E_3 = \begin{bmatrix} 1 & -1/2 \\ 0 & 1 \end{bmatrix}$

29. Given matrices $A = \begin{bmatrix} 5 & 2 \\ 3 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$, find matrices X and Y such that $AX = B$ and $YA = B$. [Hint: Both A and B are invertible.]

Answer: $X = \begin{bmatrix} -1 & 0 \\ 4 & 2 \end{bmatrix}$, $Y = \begin{bmatrix} 9 & -14 \\ 5 & -8 \end{bmatrix}$

30. Show that if A is an upper triangular invertible matrix, then so is A^{-1}.

31. Show that if A is a lower triangular invertible matrix, then so is A^{-1}.

32. Can every square matrix be written as a sum of two invertible matrices? **Answer:** Yes (over $\mathbb{R}$ and $\mathbb{C}$)

33. Can every invertible matrix be written as a sum of two non-invertible matrices? **Answer:** Yes

21. Consider the row vectors $v_1 = [1, 2, 3, 4]$, $v_2 = [2, 0, 1, 1]$, $v_3 = [-3, 2, 1, 2]$, and $v_4 = [1, -2, -2, -3]$. Construct a row vector $v \in \mathbb{R}^{4\times 1}$ which is not expressible as $av_1 + bv_2 + cv_3 + dv_4$ for any $a, b, c, d \in \mathbb{R}$. [Hint: Compute the RREF of $A = [v_1^t \;\; v_2^t \;\; v_3^t \;\; v_4^t]$.]

RREF of $A^T = \begin{bmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ $\quad A^T = \begin{bmatrix} 1 & 2 & -3 & 1 \\ 2 & 0 & 2 & -2 \\ 3 & 1 & 1 & -2 \\ 4 & 1 & 2 & -3 \end{bmatrix}$

$u_3 = u_1 - 2u_2 \qquad u_4 = -u_1 + u_2$

$u \notin \{au_1 + bu_2 + cu_3 + du_4 \; ; \; a,b,c,d \in \mathbb{R}\}$

$u \notin \{u_1(a + c - d) + u_2(b - 2c + d) \; ; \; a,b,c,d \in \mathbb{R}\}$

$u \notin \{\alpha u_1 + \beta u_2 \; ; \; \alpha, \beta \in \mathbb{R}\}$

$u \notin \left\{ \begin{bmatrix} \alpha + 2\beta \\ 2\alpha \\ 3\alpha + \beta \\ 4\alpha + \beta \end{bmatrix} \; ; \; \alpha, \beta \in R \right\}$

$\alpha = x_2/2$

$\beta = x_1/2 - x_2/4$

$x_3 = \frac{x_1}{2} + \frac{5x_2}{4}$

$x_4 = \frac{x_1}{2} + \frac{7x_2}{4}$

$u \notin \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \; ; \; x_1, x_2, x_3, x_4 \in \mathbb{R} \right\}$

Now just guess!

